

5(a)(i)	period or $T = 1 / 5000 (= 2 \times 10^{-4} \text{ s})$	C1
	time-base setting = $1.5 \times 2 \times 10^{-4} / 6.0$ or $2 \times 10^{-4} / 4.0$ $= 5 \times 10^{-5} \text{ s cm}^{-1}$	A1
5(a)(ii)	new trace drawn with same period as original trace	B1
	new trace drawn with amplitude greater than 1.0 cm	M1
	new trace drawn with amplitude of 1.7 cm	A1
5(b)(i)	path difference (from slits to P) is zero or phase difference (between waves at P) is zero (so constructive interference)	B1
5(b)(ii)	$\lambda = ax / D$	C1
	$D = (3.6 \times 10^{-4} \times 4.0 \times 10^{-3}) / 630 \times 10^{-9}$	C1
	$= 2.3 \text{ m}$	A1
5(c)	upward sloping straight line starting from a non-zero value of x at $\lambda = 400 \text{ nm}$	B1